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MSC CoSim

Release 2025.1:

What's New

August 2025

New Capabilities and Updates

MSC CoSim 2025.1

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Release Highlights

New in MSC CoSim 2025.1:

Adams – Marc coupling
- Support for Marc estimated time increment

Marc – scFLOW & MSC Nastran-scFLOW coupling
- Minimum number of CoSim iteration loops

Other enhancements
- Thermal co-simulation for Marc-scFLOW coupling via the new scFLOW command for CoSim

Enhancements for Adams-Marc Coupling

Support for Marc estimated time increment

- **Description:**

  CoSim engine sends the new time increment (*dt_new*) to Marc and Adams. However, Marc also calculates internally an estimated *dt_est* which can be smaller than *dt_new*.

  CoSim engine now considers both values and send the lower time increment to the solvers

- **Benefits:**

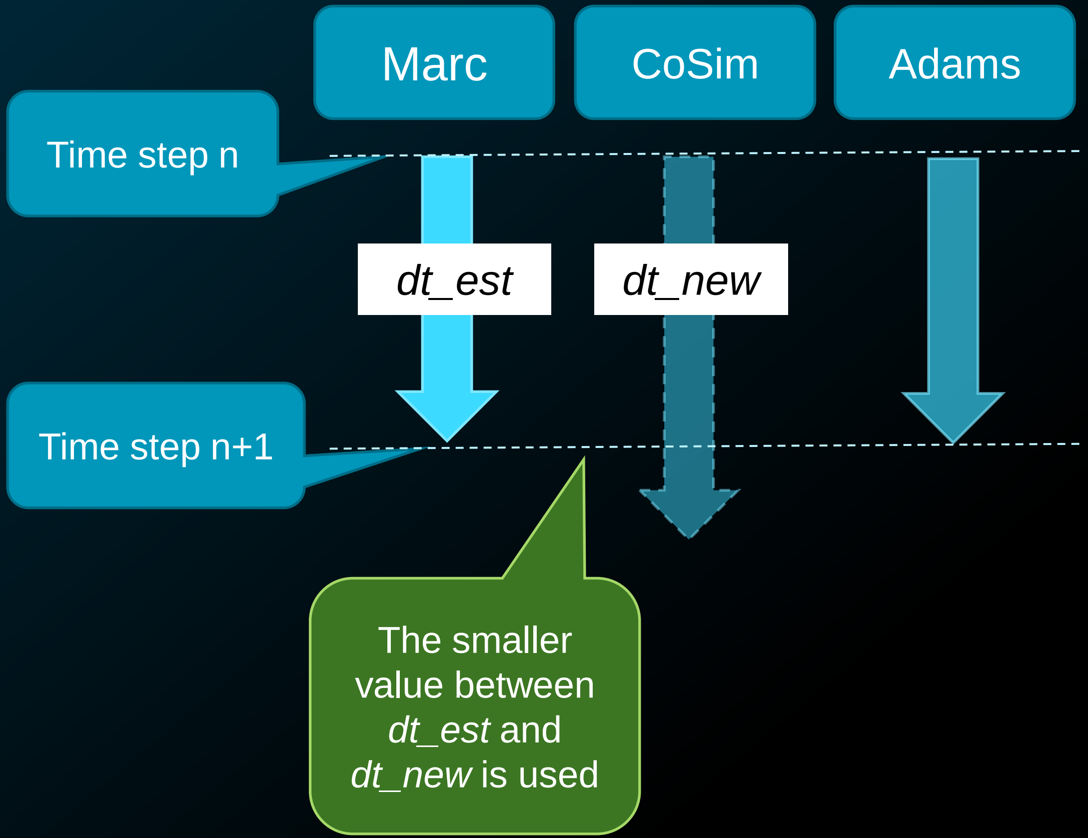
  This new criteria in the definition of the time increment has potential benefits both in performance and convergence

  The number of cutbacks can be reduced due to a more suitable time increment choice

  Any highly non-linear jobs which requires many cutbacks during the run benefit the most

- **Note:**

  Actual performance gain and cutbacks reduction vary depending on the specific problem setup and boundary conditions



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Enhancements for Marc-scFLOW & MSC Nastran-scFLOW coupling

Minimum number of CoSim iteration loops

- **Description:**

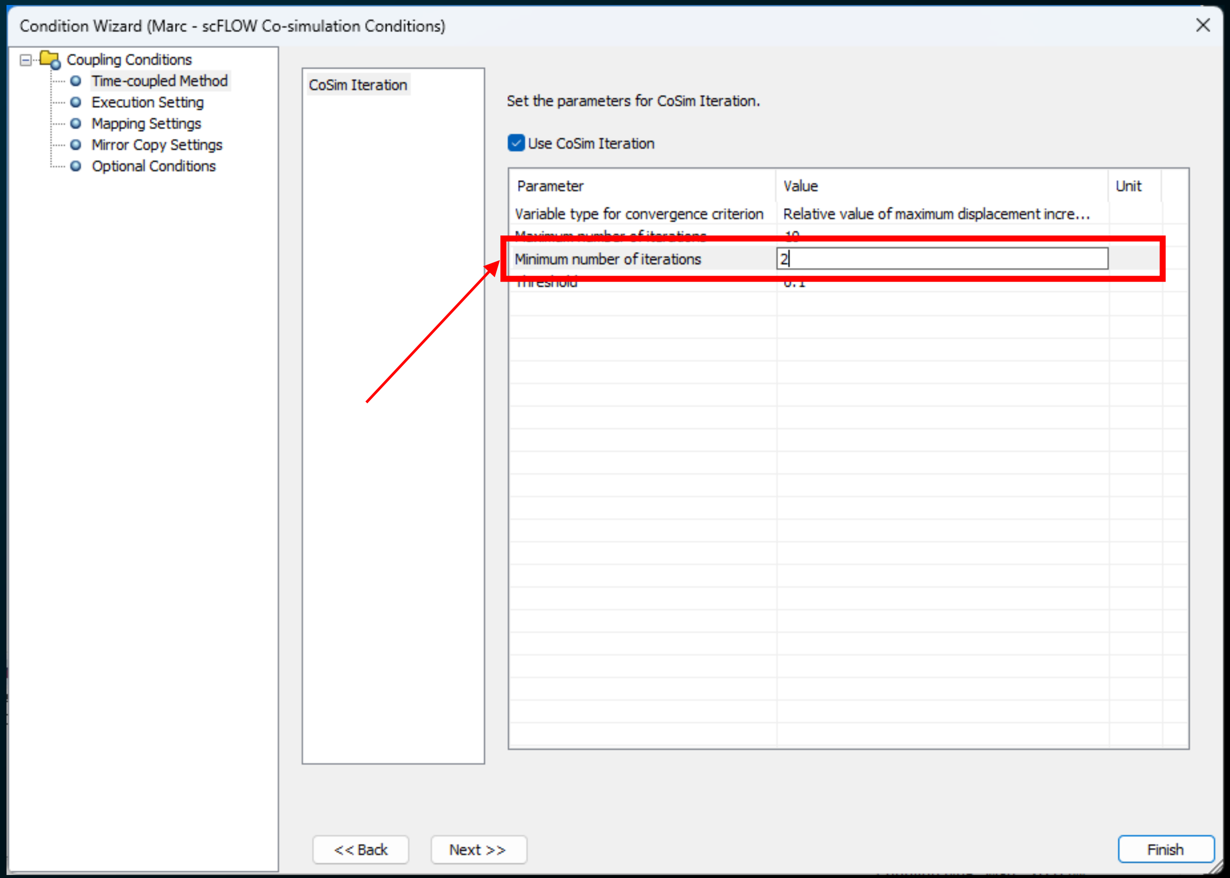
  It is now possible to set the minimum number of iterations in the iterative CoSIm mode in Marc-scFLOW and the static CoSim mode in Nastran-scFLOW

- **Benefits:**

  Improves accuracy and stability by preventing premature loop terminations

  Simplifies the setup in complex systems where convergence thresholds are hard to define

  Ideal for non-linear systems or problems hard to converge. Also, it's useful to users looking for a simple setup to achieve higher accuracy in fluid-structure co-simulations



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Other enhancements

- **Thermal co-simulation for Marc-scFLOW coupling via the new scFLOW command for CoSim**

  The thermal co-simulation in Marc-scFLOW coupling is now supported also via the new specification of the scFLOW command (COSIM command).

  (Note)

  - CoSimPRE configuration steps are the same of the conventional scFLOW command (COSIM_STRUCTURE command).
  - scFLOW pre/solver in Cradle CFD 2025.1 is required to use this feature in scFLOW.
  - The new command specification of scFLOW is available by selecting [New specification] as [Command specification for the co-simulation using MSC CoSim] in [Environment Setting] dialog in scFLOWpre.
  - Some features, such as the 6-DOF separation and restart, are only available with the new specification scFLOW command. Also, new functions related to scFLOW coupling will be added using the new command specification in the future.

Additional Resources

Help Website

[QuickAccess Links to MSC CoSim Resources (hexagon.com)]

MSC CoSim Support Home Page

2024/02/22 · Knowledge

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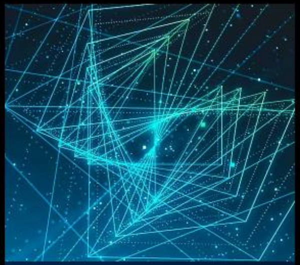

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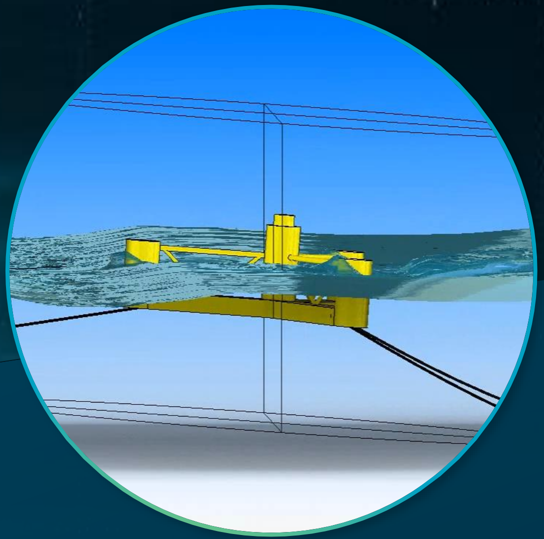

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